

Deep Learning-Based Long-Horizon MPC: Robust, High Performing, and Computationally Efficient Control for PMSM Drives

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Abstract—This article presents a computationally efficient and high performing approximate long-horizon model predictive control (MPC) for permanent magnet synchronous motors (PMSMs). Two continuous control set MPC (CCS-MPC) formulations are considered: the classical current tracking delta MPC (Del-MPC) and the torque tracking economic MPC (EMPC). To achieve offset-free torque tracking under model uncertainties and in all regions of operation, a disturbance observer and a dq -current reference generator are used. To enable real-time implementation of the long-horizon CCS-MPC, the development of a real-time capable solver is not required, since MPC approximation based on deep neural networks (DNNs) is considered and utilized for controller's evaluation at run time. The approximation is done by training the DNN to learn the MPC functionality based on offline-generated training data and in an open-loop manner. The robust and offset-free tracking performance of the proposed DNN-based approximate long-horizon Del-MPC and EMPC has been validated through simulation and real-time implementation at test bench and compared to the state-of-the-art field oriented control (FOC) using internal model controller with field-weakening (FW) part and to the exact short-horizon MPC based on the fast gradient method (FGM-MPC). Results show that the long-horizon MPC can achieve significantly faster torque transient responses in comparison with the short-horizon FGM-MPC and the conventional FOC, especially in FW region.

Index Terms—Deep learning, deep neural networks (DNNs), long horizon, model predictive control (MPC), permanent magnet synchronous motor (PMSM).

I. INTRODUCTION

PERMANENT magnet synchronous motors (PMSMs) are very attractive for drive applications, including general-purpose industrial drives, high-performance servo drives, and drives for automotive/aerospace applications. This is due to PMSMs characteristics of high torque, power density and efficiency, and their excellent dynamic response [1]. Having control techniques of PMSMs that are capable to achieve fast/accurate torque response is then of special interest.

A. State of the Art

In the literature, several well-established control schemes for PMSMs have been proposed. Although these schemes have dominated the control of high-performance drive applications during the past decades, there is still room for improvement. Field oriented control (FOC) [2] requires accurate information of the motor parameters and load conditions to achieve good drive performance. It also has varying performance based on the operating point [3]. The state-of-the-art FOC based on internal model control (IMC) method, presented in [4], has less sensitivity to machine parameters and has a simpler design procedure in comparison with conventional FOC with PI controllers. However, it still has a challenging tuning and operating-point-dependent performance. Direct torque control (DTC) [5], on the other hand, causes nonsmooth current and torque waveforms and has a variable switching frequency that can give rise to machine vibration and noise.

Model predictive control (MPC) is considered as one of the advanced control techniques that has been successfully applied in industrial applications [6]. Since the last few decades, an increasing interest has been drawn to the application of MPC in power electronics and drives due to the fast and powerful development of microprocessors. MPC, similar to other predictive control concepts, utilizes a model of the system to predict future behavior of the controlled variables. The prediction of variables is done here for a predefined horizon in time (prediction horizon) and is used to select the optimal actuation that minimizes a specific cost or objective function. Advantages of MPC include an intuitive and easy to understand concept, applicability to a variety of systems, easy consideration of multivariable case, and easy inclusion of constraints and model nonlinearities. In comparison with classical controllers, MPC has the disadvantage

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of large number of calculations and the direct influence of the model quality on the controller performance [1]. MPC for electrical drives can be classified into two groups: finite control set MPC (FCS-MPC), that takes into account the discrete switching states of the converter, and continuous control set MPC (CCS-MPC), that uses a modulator (with PWM or SVM) to realize the output control voltage. While FCS-MPC often leads to current and torque ripples and requires a relatively high and variable sampling frequency, CCS-MPC can work with larger sampling times and a fixed switching frequency, hence having a lower requirement on control architecture [7]. CCS-MPC has also a higher steady-state precision and smaller current ripples, which ensures a high performance drive control [8]. Similar to the FOC approach, CCS-MPC of PMSM is typically implemented as a field-oriented MPC (FO-MPC). The torque tracking problem is decomposed into a reference generator part, responsible for calculating current references in dq -frame corresponding to a torque reference based on offline optimization, and a current feedback MPC part, instead of conventional PI controllers, to track the reference dq -currents. An additional field-weakening (FW) part is usually used to maintain operation within voltage limit and to mitigate possible torque deviation resulting from inaccuracies and fluctuations in the machine parameters [9]. Reference currents generation is done usually in open-loop, since a torque sensor is typically not included in standard drives [10]. This, however, may represent a storage burden due to the need for storing offline-obtained optimal trajectories using lookup tables (LUTs) [11].

For the realization of MPC, several formulations of tracking MPC are proposed in the literature [12], [13]. System model augmentation using disturbance model and the delta input (δu) formulations are proposed in [12]. The incremental MPC formulation, presented in [13], is utilized for CCS-MPC of PMSM in [14]. Another approach for MPC realization is to use the concept of economic MPC (EMPC) [15]–[18], which combines economic process optimization (steady-state determination) and process control using MPC formulations with general cost functions [15]. Instead of the possibly inefficient or inappropriate simple tracking of a precalculated steady-state reference by penalizing the distance to this reference, more sophisticated economic objectives can be, directly or indirectly, encoded in the EMPC cost function [16], [17]. The authors in [10] combined both of the dq -current reference calculation and the feedback-current control parts into one compact and flexible EMPC formulation without any current tracking terms and with explicit consideration of the economic model (the nonlinear torque equation) in the cost function.

Additional computational challenges have arisen by the transition to use MPC for applications characterized by their large scale, highly nonlinear and/or fast dynamics, since the optimal control problem (OCP) must be solved repeatedly in real-time. This becomes even more challenging in case of deployment on embedded systems with constrained resources [19]. Explicit MPC approaches [20], [21] handle the large computational load by solving the OCP offline for all feasible system states to obtain the optimal control laws to be evaluated online based on the current system state [22]. Such approaches are limited, however,

to linear systems with quadratic cost function and polytopic constraints, where piecewise affine control laws can be calculated. In addition, the computational complexity grows quickly with the increase of the system's size, number of constraints, and horizon length [23]. For large-scale problems, implicit MPC methods using fast solvers and tailored implementations, like fast gradient methods (FGM) [24], [25] and the alternating direction method of multipliers (ADMM) [26], [27], are generally used to solve MPC online. These methods are also limited to linear constrained systems with quadratic cost functions and since the lower bound of the number of iterations is dependent on horizon length [24], implementation of longer horizon lengths is still a challenge.

Lately, an increasing interest has been drawn the utilization of deep learning-based techniques in approximating explicit MPC laws. The use of deep neural networks (DNNs) for approximating control laws was introduced in [23] and [28]. In contrast to conventional explicit MPC approaches, neural network approaches have a powerful approximation capability for MPC with nonlinear system models and/or nonquadratic cost functions [28]–[31], allowing an efficient online evaluation of approximated control laws, despite the relatively computationally intensive training process [32]. Research has been conducted in different aspects of DNN-approximate MPC, including bounds on required DNN's size [33], guarantees on stability and constraint satisfaction [34], and correction for steady state offset-free tracking [19]. It has also been successfully demonstrated using numerical benchmark problems and real-time practical applications [19], [33], [35]–[38]. Also in power electronics and drive control applications, neural network controllers without online learning capabilities can be adopted to approximate online-infeasible MPC control laws, like long horizon MPC, especially with the relatively high sampling frequencies in kilohertz range [39]. A prediction horizon up to three steps was considered for DNN-approximate FCS-MPC for stand-alone converters in [40]. The authors in [39] presented the control of an internal permanent magnet synchronous motor (IPMSM) using FCS-MPC algorithm with a horizon length equal to five steps, where determining the inverter switching states is treated as a classification problem that is solved offline and used to train a computationally implementable neural network, which managed to capture the performance enhancement of the long horizon FCS-MPC.

B. Contribution of This Article

Achieving efficient operation and high-performance control of electrical drives under consideration of uncertainties in all operation regions, including the FW region, is still a challenging task for the conventional FOC and DTC schemes discussed earlier. MPC-based schemes, on the other hand, have advantages in FW region through the consideration of current and voltage constraints in their formulations. However, their performance is still limited since only short prediction horizons are practically feasible at high sampling rates. Long horizons, if made possible, can help achieving significant performance improvement, enhanced stability, and robustness [41].

In this article, two long-horizon CCS-MPC formulations with offset-free torque tracking are proposed for PMSM: the classical tracking Del-MPC that is based on an online reference generator to provide the dq -current setpoint to be tracked by the MPC, and EMPC that exploits the strengths of MPC to simultaneously determine the setpoints and control the dq -currents. To achieve offset-free torque tracking under model uncertainties and in all regions of operation, including FW, a disturbance observer and a dq -current reference generator are used. The disturbance observer estimates the plant-model mismatch to be used directly in the EMPC, and in the reference generator for the determination and adjustment of dq -current setpoints required for the Del-MPC.

Despite the offset-free torque tracking capability of the proposed structure of the reference generator with Del-MPC, and the EMPC, and despite the significant performance enhancement that can be achieved using the long-horizon, especially in FW region, their online computation is infeasible at high sampling rates. Hence, to enable their real-time implementation, MPC approximation through DNN is utilized. Two real-time implementable DNNs are trained to approximate the behavior of the combined reference generator with Del-MPC, and the behavior of the EMPC using offline-generated training data. This makes the development of a real-time capable MPC solver, which is often a nontrivial or even infeasible task, not required.

To facilitate the training process, training data are generated in an open-loop manner based on the motor's nominal model (no system closed-loop simulation or measured data are required). Table I compares the horizon length and the sampling time of this article's implementation with other implementations of CCS-MPC for PMSM from the literature.

The approximate DNN-based MPC is compared to its exact counterpart (online-solved-optimization-based MPC) in simulation to show the efficient approximation capability of the DNN for long-horizon CCS-MPC. In addition to that, both DNN-based MPC formulations are implemented at the test bench and their benefit in enhancing the control performance, especially in FW region, is demonstrated in comparison to both the state-of-the-art IMC-based FOC with FW controller and to the exact (online-solved) short-horizon MPC using the fast gradient method (FGM-MPC).

The main contribution of the article is the proposition of a holistic control approach for PMSM, that is high performing as well as robust in all operation areas, especially in the challenging FW region, and at the same time real-time capable. More precisely, the contribution consists of the following key elements.

- 1) This article presents for the first time, to the best of authors' knowledge, the development and experimental validation of a neural network-based approximation of CCS-MPC with long horizons for PMSM. MPC with long horizon leads to an enhanced performance, especially in the FW region, and the usage of DNNs makes it real-time capable, and hence, suitable for real-time implementation.
- 2) An online-efficient, high performing, and robust offset-free control scheme is proposed, which is based on a tailored composition of DNN and disturbance observer. Through the proposed structure, the plant-model

mismatch is determined efficiently online using the disturbance observer and is considered as an input to the DNN, and thus, can be compensated. This implies that, the DNN can be trained based on a nominal model and thus completely offline without the need of any measurements.

- 3) The superior control performance of the proposed DNN-based approximate long-horizon Del-MPC and EMPC in comparison with state-of-the-art IMC-based FOC and the exact (online-solved) short-horizon FGM-MPC is demonstrated.

The rest of this article is organized as follows. Section II presents a brief description of the PMSM model and plant-model mismatch disturbance estimation. The proposed CCS-MPC formulations are discussed in Section III. The procedure for the DNN-based MPC approximation is presented in Section IV. Simulation analysis is found in Section V and the experimental results are discussed in Section VI. Finally, Section VII concludes this article.

II. MODELING OF PMSM

The continuous-time model of the PMSM in the synchronously rotating dq -frame can be expressed as follows [9]:

$$\underbrace{\begin{bmatrix} \frac{di_d(t)}{dt} \\ \frac{di_q(t)}{dt} \end{bmatrix}}_{\mathbf{\dot{x}}} = \underbrace{\begin{bmatrix} -\frac{R_s}{L_d} & \omega_r \frac{L_q}{L_d} \\ -\omega_r \frac{L_d}{L_q} & -\frac{R_s}{L_q} \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} i_d(t) \\ i_q(t) \end{bmatrix}}_{\mathbf{x}} + \underbrace{\begin{bmatrix} \frac{1}{L_d} & 0 \\ 0 & \frac{1}{L_q} \end{bmatrix}}_{\mathbf{B}} \underbrace{\begin{bmatrix} u_d(t) \\ u_q(t) \end{bmatrix}}_{\mathbf{u}} + \underbrace{\begin{bmatrix} 0 \\ -\omega_r \frac{\psi_{pm}}{L_q} \end{bmatrix}}_{\mathbf{f}} \quad (1)$$

where $\mathbf{x}$ is the state vector, $\mathbf{u}$ is the input vector, $\mathbf{A}$ is the system matrix, $\mathbf{B}$ is the input matrix, $\mathbf{f}$ is a known disturbance vector, $i_d(t)$, $i_q(t)$, $u_d(t)$, $u_q(t)$, L_d , and L_q are the d - and q -axes components of the stator current (in A), voltage (in V), and inductance (in H) of the PMSM, respectively, R_s is the stator resistance (in Ω), ω_r is the electrical angular speed of the rotor flux (in rad / s), and ψ_{pm} is the flux linkage of the rotor's permanent magnet (in V · s). The electromagnetic torque $T_e(t)$ (in N · m) is expressed in the dq -frame as

$$T_e(t) = \frac{3}{2} Z_p [(L_d - L_q) i_d(t) i_q(t) + i_q(t) \psi_{pm}] \quad (2)$$

where Z_p is the number of pole pairs.

For MPC design and implementation on digital processors, discretization using forward Euler (explicit) method with sampling time T_s (in s) is applied to the continuous-time model (1) to obtain the discrete-time model of the PMSM

$$\mathbf{x}_{k+1} = \mathbf{A}_d \mathbf{x}_k + \mathbf{B}_d \mathbf{u}_k + \mathbf{f}_d \quad (3)$$

where k is the current sampling instant, $\mathbf{x}_k = [i_{d_k}, i_{q_k}]^\top$, $\mathbf{u}_k = [u_{d_k}, u_{q_k}]^\top$, $\mathbf{A}_d = T_s \mathbf{A} + \mathbf{I}$, $\mathbf{B}_d = T_s \mathbf{B}$, and $\mathbf{f}_d = T_s \mathbf{f}$.

Plant-model mismatch due to model parameter inaccuracy is considered as input disturbance in this article and it is crucial for achieving offset-free tracking as shown in the next section. Due to the much slower rate of change of the disturbance in

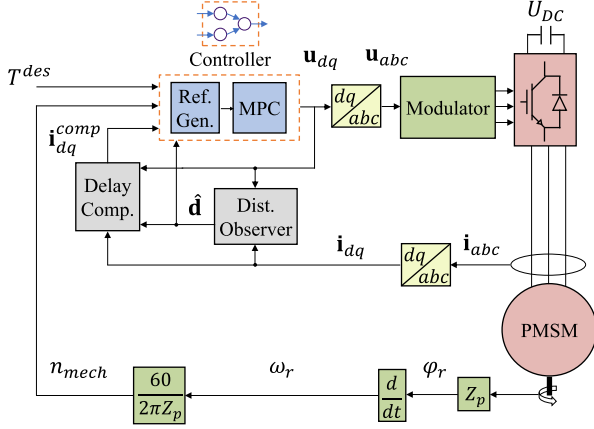


Fig. 1. FO-MPC scheme for PMSM.

comparison with the system's states, the disturbance is assumed constant in the dq -frame. A disturbance estimator is used to determine an estimation of the input disturbance at run time. Different observer concepts can be used for parameter identification and disturbance estimation of the PMSM, e.g., Luenberger observer [[47]], Kalman filter [48], [49], sliding mode observer [50], and nonlinear observer [51]. For its ease of implementation and reasonable computational burden, the Kalman filter is considered in this article. Therefore, a discrete-time Kalman filter implementation is applied utilizing the discrete-time augmented system and disturbance model

$$\begin{bmatrix} \mathbf{x}_{k+1} \\ \mathbf{d}_{k+1} \end{bmatrix} = \begin{bmatrix} \mathbf{A}_d & \mathbf{B}_d \\ \mathbf{0}_{2 \times 2} & \mathbf{I}_{2 \times 2} \end{bmatrix} \begin{bmatrix} \mathbf{x}_k \\ \mathbf{d}_k \end{bmatrix} + \begin{bmatrix} \mathbf{B}_d \\ \mathbf{0}_{2 \times 2} \end{bmatrix} \mathbf{u}_k + \begin{bmatrix} \mathbf{f}_d \\ \mathbf{0}_{2 \times 1} \end{bmatrix} \quad (4a)$$

$$\mathbf{y}_k = \begin{bmatrix} \mathbf{I}_{2 \times 2} & \mathbf{0}_{2 \times 2} \end{bmatrix} \begin{bmatrix} \mathbf{x}_k \\ \mathbf{d}_k \end{bmatrix}. \quad (4b)$$

The estimation $\hat{\mathbf{X}}_k = [\hat{\mathbf{x}}_k, \hat{\mathbf{d}}_k]^\top$ contains both the motor states (dq -currents) and the plant-model mismatch $\hat{\mathbf{d}}_k = [\hat{d}_{d_k}, \hat{d}_{q_k}]^\top$ as a dq -input voltage disturbance. For more information on the concept and the implementation, it is referred to [52] and [53].

III. MPC FOR PMSM

The considered FO-MPC scheme for the PMSM is shown in Fig. 1. Transformed measured stator's phase currents are fed, after compensating controller delay, into the controller block together with the measured mechanical speed (n_{mech}), desired torque (T^{des}), and estimated plant-model mismatch disturbance ($\hat{\mathbf{d}}$). The MPC outputs the control dq -input voltage ($\mathbf{u}_{dq}$) that is sent, after back transformation to the abc -frame, to the modulator, which is then realized at the motor's terminals by the inverter.

As mentioned earlier, real-time implementation of the online reference generator and the long-horizon CCS-MPC is to be made possible through DNN approximation. The trained DNN takes the same controller inputs and approximates the control dq -voltage to be applied at the motor, as shown in Fig. 2.

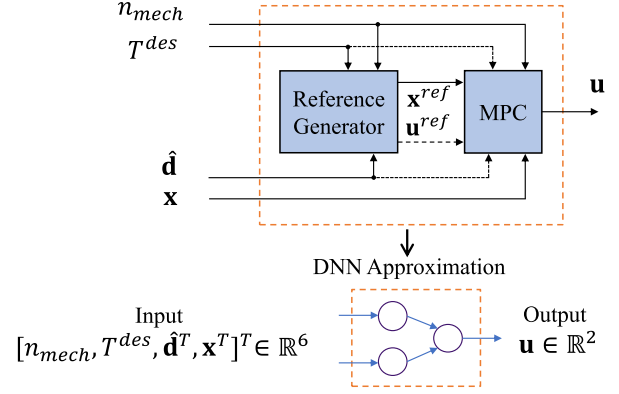


Fig. 2. DNN-based MPC.

Torque tracking of PMSM is considered here using two advanced MPC formulations: offset-free current tracking MPC and economic MPC.

A. Offset-Free Current Tracking MPC

Two aspects need to be considered in order to achieve an offset-free torque response of the PMSM. The first one is related to the generation of correct reference dq -currents that correspond to the desired torque, since plant-model mismatch and parameters variation during operation can cause the reference currents calculated based on offline optimization to result in torque deviation, especially when operating at voltage limit. This issue should be considered in the design of the reference generator. The second aspect is related to the current feedback controller (current tracking MPC) and its capability to steer the currents of the machine to the provided reference dq -currents without steady-state deviation.

To avoid the storage burden and inaccuracies associated with offline-calculated reference currents trajectories based on maximum torque per ampere (MTPA) and maximum torque per volt (MTPV) criteria, and to avoid the need for additional field-weakening controller with its complicated tuning and its dynamic negative effect on the fast response capability of MPC, a generic online reference generator is designed for calculating the steady-state reference dq -current $\mathbf{x}^{ref} = [i_d^{ref}, i_q^{ref}]^\top$ and dq -voltage $\mathbf{u}^{ref} = [u_d^{ref}, u_q^{ref}]^\top$ corresponding to a desired torque T^{des} by solving the following optimization problem:

$$\underset{T^{ref}, \mathbf{x}^{ref}, \mathbf{u}^{ref}}{\text{minimize}} \quad J := \|\mathbf{x}^{ref}\|^2 + w \cdot (T^{des} - T^{ref})^2 \quad (5a)$$

$$\text{subject to} \quad \mathbf{0} = (\mathbf{A}_d - \mathbf{I})\mathbf{x}^{ref} + \mathbf{B}_d(\mathbf{u}^{ref} + \hat{\mathbf{d}}) + \mathbf{f}_d \quad (5b)$$

$$T^{ref} = \frac{3}{2} Z_p [(L_d - L_q) i_d^{ref} i_q^{ref} + i_q^{ref} \psi_{pm}] \quad (5c)$$

$$\|\mathbf{x}^{ref}\|^2 \leq I_{dq_{max}}^2 \quad (5d)$$

$$\|\mathbf{u}^{ref}\|^2 \leq U_{dq_{max}}^2. \quad (5e)$$

The objective function J in (5a) consists of a current minimization term for copper loss reduction and a quadratic torque

tracking (or torque error) term, with weight w due to the different nature of both terms, to minimize the difference between reference and desired torque. Equality constraint (5b) represents the discrete-time model of the PMSM with estimated plant-model mismatch under the steady-state condition ($\mathbf{x}_{k+1} = \mathbf{x}_k = \mathbf{x}^{\text{ref}}$ with $\mathbf{u}_k = \mathbf{u}^{\text{ref}}$). Equation (5c) is the motor's torque equation (2) to relate reference torque to reference current. Inequality constraint (5d) is the dq -current constraint with $\|\mathbf{x}^{\text{ref}}\|^2 = \mathbf{x}^{\text{ref}\top} \mathbf{x}^{\text{ref}}$ and $I_{dq_{\max}} = \sqrt{2}I_{\max}$, where $I_{\max}$ is the motor maximum rms current. Inequality constraint (5e) is the constraint on dq -voltage, where $U_{dq_{\max}} = U_{dc}/\sqrt{3}$ is the maximum continuous amplitude of the fundamental stator voltage (inner circle inscribed in the voltage hexagon of the two-level inverter) in relation to the inverter's dc bus voltage limitation.

The minimization of copper loss in (5a) will ensure an operation according to MTPA criteria, when applicable, and the direct handling of current and voltage constraints in (5d) and (5e) will automatically consider other regions of operation. Inclusion of T^{ref} as an optimization variable and the use of quadratic torque error term in the objective function ensure existence of a solution even in case of infeasible desired torque. In such a case, it will compute the closest possible feasible T^{ref} to T^{des} , with its corresponding $\mathbf{x}^{\text{ref}}$ and $\mathbf{u}^{\text{ref}}$. Considering $\hat{\mathbf{d}}$ in the steady-state equation (5b) accounts for model inaccuracies and adjusts the reference to maintain voltage limit, without the need for any field-weakening controllers, to achieve offset-free torque tracking. Furthermore, in case of online estimated machine parameters ($\hat{L}_d, \hat{L}_q, \hat{R}_s, \hat{\psi}_{pm}$), this form of the reference generator can easily consider them, allowing a possible and easy extension to adaptive control.

For the design of an offset-free model predictive current controller capable of tracking reference currents, generated from the described reference generator, without steady-state deviation, a simple tracking MPC formulation is considered. This delta formulation (Del-MPC) depends on the deviation from the steady-state references as $\Delta\mathbf{x}_k = \mathbf{x}_k - \mathbf{x}^{\text{ref}}$ and $\Delta\mathbf{u}_k = \mathbf{u}_k - \mathbf{u}^{\text{ref}}$, where $\mathbf{x}^{\text{ref}}$ and $\mathbf{u}^{\text{ref}}$ are obtained from solving the reference generator optimization problem (5). This is much simpler than the formulations proposed in [12]–[14], which complicate the MPC design and require a state observer and extra inputs.

Based on this definition, the current tracking Del-MPC is formulated as

$$\underset{\Delta\mathbf{U}, \Delta\mathbf{X}}{\text{minimize}} \quad \|\Delta\mathbf{x}_N\|_P^2 + \sum_{k=0}^{N-1} \|\Delta\mathbf{x}_k\|_Q^2 + \|\Delta\mathbf{u}_k\|_R^2 \quad (6a)$$

$$\text{subject to} \quad \Delta\mathbf{x}_0 = \mathbf{x} - \mathbf{x}^{\text{ref}}, \quad (6b)$$

$$\Delta\mathbf{x}_{k+1} = \mathbf{A}_d \Delta\mathbf{x}_k + \mathbf{B}_d \Delta\mathbf{u}_k \quad k = 0, \dots, N-1, \quad (6c)$$

$$\|\Delta\mathbf{x}_k + \mathbf{x}^{\text{ref}}\|^2 \leq I_{dq_{\max}}^2 \quad k = 1, \dots, N, \quad (6d)$$

$$\|\Delta\mathbf{u}_k + \mathbf{u}^{\text{ref}}\|^2 \leq U_{dq_{\max}}^2 \quad k = 0, \dots, N-1 \quad (6e)$$

where $\Delta\mathbf{U} = [\Delta\mathbf{u}_0^\top, \dots, \Delta\mathbf{u}_{N-1}^\top]^\top$ and $\Delta\mathbf{X} = [\Delta\mathbf{x}_0^\top, \dots, \Delta\mathbf{x}_N^\top]^\top$. The applied control input is $\mathbf{u} = \Delta\mathbf{u}_0^* + \mathbf{u}^{\text{ref}}$, where $\Delta\mathbf{u}_0^*$ is the first element of the optimal control sequence $\Delta\mathbf{U}^*$, which is the minimizer of the optimization problem. In the cost function (6a), stage cost weighing matrices are $\mathbf{Q} \succeq 0$ and $\mathbf{R} \succ 0$. $\mathbf{P}$ is the terminal cost weighing matrix and satisfies the Riccati equation $\mathbf{P} = \mathbf{A}_d^\top \mathbf{P} \mathbf{A}_d - (\mathbf{A}_d^\top \mathbf{P} \mathbf{B}_d)(\mathbf{B}_d^\top \mathbf{P} \mathbf{B}_d + \mathbf{R})^{-1}(\mathbf{B}_d^\top \mathbf{P} \mathbf{A}_d) + \mathbf{Q}$ [12]. The matrices $\mathbf{A}_d$ and $\mathbf{B}_d$ in (6c) correspond to the discrete-time PMSM model (3) and $\mathbf{x}$ is the known (measured and compensated) system states. Note that $k=0$ is omitted from the state constraints in (6d) to mitigate the infeasibility that may occur in case of not feasible measured initial states $\|\mathbf{x}\|^2 > I_{dq_{\max}}^2$. One can note also that the known disturbance vector $\mathbf{f}_d$, which is part of the discrete-time model (3), does not appear in (6c). In addition, the estimated plant-model mismatch input disturbance $\hat{\mathbf{d}}$ is not considered in (6). This is due to the use of $\Delta\mathbf{x}_k$ and $\Delta\mathbf{u}_k$ instead of $\mathbf{x}_k$ and $\mathbf{u}_k$ in the MPC formulation.

B. Economic MPC

Separating the task of determining the optimal steady-state references of the controlled system, corresponding to desired objectives, from the task of guiding the system's transient states to their references and maintaining them at their references under dynamic disturbances, by penalizing the distance to these references, does not result necessarily in the optimal response. Many large-scale controlled systems have economic objectives, like operation efficiency, cost, etc., where this hierarchal separation can be inefficient or inappropriate [17]. In control of chemical processes, for example, operation at a steady-state reference is not necessarily the economically optimal operation. EMPC solves this by replacing the simple tracking stage cost function with a general cost function that is related, directly or indirectly, to the economic nature of the controlled system [15]. For the torque control of the PMSM, explicit consideration of the economic model (the nonlinear torque equation) in the OCP instead of simply tracking the precalculated reference dq -currents can be an option. Based on [10], EMPC can be formulated as

$$\underset{\mathbf{U}, \mathbf{X}, \mathbf{s}}{\text{minimize}} \quad \beta \cdot \|\mathbf{x}_N - \mathbf{x}^{\text{ref}}\|^2 + \sum_{k=0}^{N-1} \|\mathbf{x}_k\|^2 + w \cdot s_k \quad (7a)$$

$$\text{subject to} \quad \mathbf{x}_0 = \mathbf{x} \quad (7b)$$

$$\mathbf{x}_{k+1} = \mathbf{A}_d \mathbf{x}_k + \mathbf{B}_d (\mathbf{u}_k + \hat{\mathbf{d}}) + \mathbf{f}_d \quad k = 0, \dots, N-1 \quad (7c)$$

$$T(\mathbf{x}_k) = \frac{3}{2} Z_p [(L_d - L_q) i_{d_k} i_{q_k} + i_{q_k} \psi_{pm}] \quad k = 0, \dots, N \quad (7d)$$

$$-s_k \leq T(\mathbf{x}_k) - T^{\text{des}} \leq s_k \quad k = 0, \dots, N \quad (7e)$$

$$s_k \geq 0 \quad k = 0, \dots, N \quad (7f)$$

$$\|\mathbf{x}_k\|^2 \leq I_{dq_{\max}}^2 \quad k = 1, \dots, N \quad (7g)$$

$$\|\mathbf{u}_k\|^2 \leq U_{dq_{\max}}^2 \quad k = 0, \dots, N-1 \quad (7h)$$

$$\|\mathbf{B}_d^{-1}[(\mathbf{A}_d - \mathbf{I})\mathbf{x}_N + \mathbf{B}_d\hat{\mathbf{d}} + \mathbf{f}_d]\|^2 \leq U_{dq_{\max}}^2 \quad (7i)$$

where $\mathbf{U} = [\mathbf{u}_0^\top, \dots, \mathbf{u}_{N-1}^\top]^\top$ is the control input sequence, $\mathbf{X} = [\mathbf{x}_0^\top, \dots, \mathbf{x}_N^\top]^\top$ is the states sequence, and $\mathbf{s} = [s_0, \dots, s_N]^\top$ is the slack variable sequence. The torque $T(\mathbf{x}_k)$ is calculated from (2) and $\hat{\mathbf{d}}$ is the estimated disturbance at $k = 0$ and is assumed constant throughout the prediction horizon ($\hat{\mathbf{d}}_k = \hat{\mathbf{d}}_0 = \hat{\mathbf{d}}, k = 1, \dots, N-1$). The terminal set (7i) represents the set of feasible steady-states, bounded by an ellipse that depends on the rotor speed ω_r , and is used to avoid convergence and stability problems [10], [54].

In comparison to the tracking Del-MPC formulation, the current and voltage tracking stage cost is replaced by a current minimization term and a slack variable term to enforce the desired torque at minimum phase current. A terminal cost that penalizes the deviation between the last predicted state $\mathbf{x}_N$ and the steady-state reference $\mathbf{x}^{\text{ref}}$, computed from (5), is used. This was motivated by [55], where the use of a steady-state terminal equality constraint was proposed for proving stability of EMPC with a Lyapunov function. It can also be noted, that in contrast to the tracking Del-MPC formulation in (6), the known disturbance vector $\mathbf{f}_d$ and the estimated plant-model mismatch $\hat{\mathbf{d}}$ appear explicitly in the EMPC formulation. This is due to the use of absolute states $\mathbf{x}_k$ and input $\mathbf{u}_k$ as the MPC variables.

So, by considering both the reference generator problem (5) and the MPC problem, (6) or (7), the two sequential problems can be expressed as $\mathbb{P}_N(n_{\text{mech}}, T^{\text{des}}, \hat{\mathbf{d}}, \mathbf{x})$, where n_{mech} is required as input since the matrix $\mathbf{A}_d$ and the vector $\mathbf{f}_d$ in (5b), (6c), and (7c) depend on ω_r that can be calculated from n_{mech} and the number of pole pairs Z_p as $\omega_r = Z_p n_{\text{mech}} \cdot \frac{2\pi}{60}$. The inputs T^{des} and $\hat{\mathbf{d}}$ are required for the reference generator (5) to calculate $\mathbf{x}^{\text{ref}}$ and $\mathbf{u}^{\text{ref}}$, and for EMPC (7). Initialization of MPC is done with the measured states $\mathbf{x}$ in (6b) and (7b). The remaining parameters are considered constant.

By solving $\mathbb{P}_N$, the control input $\mathbf{u}$ is obtained and sent, after back transformation to the abc-frame, to the modulator. As will be discussed in the next section, a DNN is used to approximate this solution, to enable its real-time implementation, and hence replacing both the reference generator and the MPC, as shown in Fig. 2.

It should be noted that, comparing the performance of both CCS-MPC approaches discussed above for PMSM, has not been done so far in the literature. This has been considered in this article, along with the influence of horizon length on their performance, both in simulation and in experiments. The proposed long-horizon MPC formulations, including the reference generator, are not intended to be solved online. Their offline solution, obtained using any existing solver, is needed for the training of the DNNs that are going to be evaluated at run time as shown in the next section. This approach enables real-time implementation of long-horizon MPC and makes the development of a real-time capable solver obsolete.

IV. DEEP LEARNING-BASED APPROXIMATE MPC

A. Approximate MPC Using DNN

Neural networks, first introduced in [56], were inspired by a similitude to the information processing and communication nodes of the brain. A neural network is a nonlinear parametric function that maps a set of input variables into a set of output variables according to adjustable parameters. A feed-forward deep neural network, with L hidden layers and M neurons per hidden layer, can be written as

$$\mathbf{y} = g(\mathbf{z}, \Theta) = f_{L+1} \circ h_L \circ f_L \circ \dots \circ h_1 \circ f_1(\mathbf{z}) \quad (8)$$

where $\mathbf{z} \in \mathbb{R}^{n_z}$ is the network's input, $\mathbf{y} \in \mathbb{R}^{n_y}$ is the network's output, and the adjustable (to be optimized) parameters $\Theta = \{\theta_1 \dots \theta_{L+1}\}$ with $\theta_l = \{\mathbf{W}_l, \mathbf{b}_l\} \quad \forall l = 1, \dots, L+1$.

The weights are $\mathbf{W}_l \in \mathbb{R}^{\alpha \times \beta}$, where $(\alpha, \beta) = (M, n_z)$ for $l = 1$, $(\alpha, \beta) = (M, M)$ for $l = 2, \dots, L$, and $(\alpha, \beta) = (n_y, M)$ for $l = L+1$. The biases are $\mathbf{b}_l \in \mathbb{R}^\gamma$, where $\gamma = M$ for $l = 1, \dots, L$ and $\gamma = n_y$ for $l = L+1$.

The affine functions defining each layer l

$$f_l(\zeta_{l-1}) = \mathbf{W}_l \zeta_{l-1} + \mathbf{b}_l \quad (9)$$

where $\zeta_{l-1} \in \mathbb{R}^M$ is the output of the previous layer and $\zeta_0 = \mathbf{z}$, are mapped using a differentiable element-wise nonlinear activation function $h_l(\cdot)$ to form the layer's output

$$\zeta_l = h_l \circ f_l(\zeta_{l-1}). \quad (10)$$

The optimal DNN parameters Θ^* , defining a trained DNN, are found by minimizing a loss function, e.g., the mean square error (MSE)

$$\Theta^* = \underset{\Theta}{\operatorname{argmin}} \frac{1}{N_s} \sum_{i=1}^{N_s} \|\mathbf{y}_i - g(\mathbf{z}_i, \Theta)\|^2 \quad (11)$$

using a training dataset consisting of a sequence of N_s samples of inputs $\mathbf{z}_i$ and their corresponding outputs $\mathbf{y}_i$.

As mentioned before, the main idea in approximate MPC is to represent mapping of the controller input to its output, i.e., the optimal solution of the OCP, by a DNN. Therefore, the following procedure is applied [19].

- 1) The feasible augmented input space $Z \subset \mathbb{R}^{n_{\text{mech}}} \times \mathbb{R}^{n_{T^{\text{des}}}} \times \mathbb{R}^{n_{\hat{\mathbf{d}}}} \times \mathbb{R}^{n_{\mathbf{x}}}$ is sampled to generate N_s samples: $\{d_i = (n_{\text{mech}_i}, T_i^{\text{des}}, \hat{\mathbf{d}}_i^\top, \mathbf{x}_i^\top)\}_{i=1}^{N_s}$.
- 2) For every sample d_i , the OCP $\mathbb{P}_N(d_i)$ is solved to obtain the exact control law $\mathbf{u}_i^* = \mathcal{K}_N(d_i)$, which is combined with d_i to form the input-output pair $(d_i^\top, \mathbf{u}_i^{*\top})$.
- 3) The dataset $\mathcal{D} = \{(d_i^\top, \mathbf{u}_i^{*\top})\}_{i=1}^{N_s}$ is used to train the DNN, i.e., finding a parametrized function $\mathcal{K}_{\text{approx}}(\mathbf{z}, \Theta)$ as the approximate MPC law.

In typical deep learning applications, like computer vision or speech recognition, training data are usually available and the focus is on designing the DNN architecture and the learning technique to meet the application requirements. The use of DNN for approximating the MPC laws requires, however, special treatment. Besides deciding on a suitable DNN structure, training data are not available and need to be generated by a suitable sampling of the exact (ground truth) MPC.

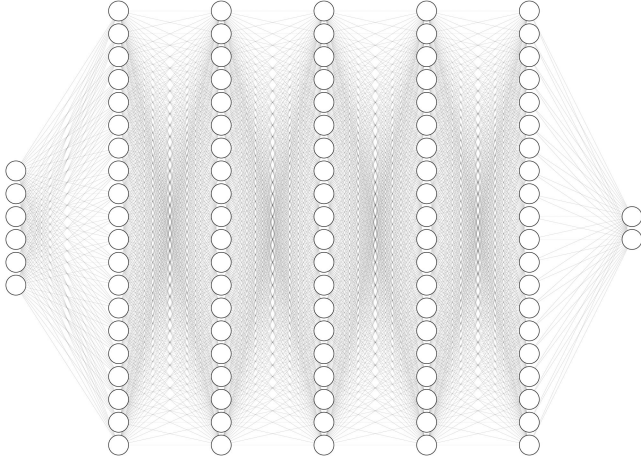


Fig. 3. Schematic of the selected DNN with $(L = 5, M = 20)$.

B. DNN Architecture

A neural network with rectified linear unit (ReLU) activation functions is a piecewise affine function, where according to [[33], Lemma 1], the number of affine regions a deep neural network can approximate depends mainly on the network's depth L , given that the network's width M is not smaller than the number of inputs (n_z). A further slight increase in the number of affine regions can be achieved if $(M \geq 3n_z)$. The MPC problem to be approximated here has $n_z = 6$ inputs. So, a network's width of $M = 20$ neurons is considered. Preliminary investigations showed that a depth more than $L = 5$ layers results in no significant improvement for the intended application while increasing the training time significantly. So, the network's depth is set to $L = 5$ layers. ReLU activation function is considered, since it is suitable for representing explicit MPC laws according to [33]. In addition to that, ReLU represents a simple function (clipping) which makes it suitable for real-time implementation.

Since obtaining the most accurate DNN for approximating the MPC laws is not the aim of this article, the selection of DNN structure is based mainly on the sampling time constraint of the embedded system on which the DNN-based approximate MPC is to be employed. The DNN with $(L = 5, M = 20)$ and ReLU) has a running time around $12 \mu\text{s}$, based on measured running time on the rapid prototyping platform dSPACE, and hence, represents a feasible choice for the used sampling time of $T_s = 100 \mu\text{s}$. A schematic of the chosen DNN architecture is shown in Fig. 3.

C. Training Data Selection

Training data samples should be chosen to reflect the characteristic of the control application by covering the whole span of operating points that are expected during online operation. For the application here, performing closed-loop simulations of the whole system over a number of time steps starting from a random specified set of initial states ($\mathbf{z}(0) \in \mathbb{R}^6$), as described in [19], can be time consuming and also challenging to cover the whole span of possible operating points. The open-loop strategy, which

evaluates the exact control law at an explicitly specified set of possible inputs, obtained by equidistant gridding, and then combining them to form the augmented input space $Z \subset \mathbb{R}^6$ can result in unnecessarily large number of samples that include highly unlikely operating points. Another way is to use random sampling, where a predefined number of samples (determined by trial and error according to the size of the problem) is used to generate sample points, where each input is randomly chosen (under uniform distribution) from the set $[x_{\min}, x_{\max}] \subset \mathbb{R}$ with x representing each one of the six inputs. This is found to be a simpler way to capture the behavior of the controller function to be approximated with much less number of samples. The limits $x_{\min}$ and $x_{\max}$ can also be relaxed to include values that are unlikely to appear during normal operation but without increasing the number of samples much as in the equidistant gridding. Including such points in the training samples can be beneficial to assure that the trained DNN will be able to deal with any limits violation that may occur, for whatever reason, during online operation (like current limit violation for example).

Table II shows the training, validation, and testing performance (as mean square error) for the case of approximate EMPC as an example. Two DNN sizes were investigated with two sets of data: a set with 500 000 random samples and a set with one million random samples. Datasets were divided as 70% for training, 20% for validation, and 10% for testing. It can be noted that for each single scenario, training, validation, and testing errors are consistent, meaning there is no overfitting. Also, small and large datasets have comparable performance, meaning there is no under fitting. So, the 500 000 samples are enough to capture the controller behavior, and the considered DNN with $(L = 5, M = 20)$ is expected to have a good approximation performance.

Although the random sampling strategy is found enough to capture the controller behavior during transient, there is still a considerable approximation error at some steady-state operating points. Having higher accuracy at steady-state is more important than in transient, since the system is expected to spend most of its operating time at steady-state. So, more focus on steady-state should be considered in generating the training samples. For this purpose, an additional dataset is generated by equidistant gridding of the inputs ($n_{\text{mech}}, T^{\text{des}}, \hat{\mathbf{d}}$) without considering the dq -currents $\mathbf{x}$. For each extra sample, the corresponding steady-state reference currents (i_d^{ref} and i_q^{ref}) are calculated using the reference generator and used, with 8 additional current combinations in a close vicinity of $(i_d^{\text{ref}}, i_q^{\text{ref}})$, to form a set of input samples with focus only on current values around the steady-state operating point. The use of few values for the current instead of equidistant grids results in significant decrease in the number of samples in comparison with equidistant gridding of all inputs.

By combining the steady-state focused set with the random set, a training set $\mathcal{D} = \{(d_i^{\text{ref}}, \mathbf{u}_i^{\text{ref}})\}_{i=1}^{N_s}$ of approximately 10^6 samples covering the whole span of the operating points with focus on steady-state points is formed and used for DNN training.

TABLE I

THIS ARTICLE'S IMPLEMENTATION IN COMPARISON WITH OTHER IMPLEMENTATIONS OF CCS-MPC FOR PMSM IN LITERATURE

Reference	Control/prediction horizon	Sampling time
[42]	5	500 μ s
[43], [44]	10	500 μ s
[45]	$N_c = 1, N_p = 3$	300 μ s
[10]	1	250 μ s
[46]	$N_c = 5, N_p = 20$	200 μ s
[14]	2	100 μ s
Article implementation	20	100 μ s

It is worth mentioning here that no measured data are needed. Only data generated offline in an open-loop manner, based on the nominal motor model, is used for the training of the DNN. This facilitates and accelerates the design process of the controller.

The training procedure, steps 1–3 discussed above, can be evaluated as follows. Step 1 (sampling) is a simple and not computationally demanding procedure (few minutes). In step 2, the OCP has to be solved N_s times. These OCPs are, however, independent from each other and can be solved in parallel. For the DNN training in step 3, many efficient tool boxes exist which exploit parallel computations. For the implementation in this article, MATLAB's Neural Network Toolbox was used and the total required learning time (including sampling, OCP solving and DNN training) is around 1 d on a personal computer. It should be noted that the training procedure can be fully automated and the development of a real-time solver, which is a nontrivial task for long horizons, is not required.

V. SIMULATION ANALYSIS

A. Simulation Setup

The proposed MPC formulations are first tuned and validated with the help of the simulation model of the test bench motor. A detailed simulation model in MATLAB/Simulink, including the motor, inverter, and controller, as shown in Fig. 1, is considered. The low-cost traction motor considered in this article is a 4-kW interior permanent magnet synchronous motor (IPMSM), presented in [10], fed from a two-level 3-phase inverter with a switching frequency of 10 kHz, which is also considered as sampling frequency of the controller. For implementation of the exact MPC algorithms in simulation, the open-source software package CasADi with the interior point solver (IPOPT) [57], was used as a rapid and efficient nonlinear programming solver (offline). The DNN training, based on Levenberg–Marquardt backpropagation algorithm, and implementation were done with the help of MATLAB's Neural Network Toolbox. Table III lists the Frobenius norm of the weight matrices for both trained DNNs corresponding to Del-MPC and EMPC. The flowchart in Fig. 4 shows how the DNN is implemented to obtain the approximated MPC output.

B. Prediction Horizon Length

An important parameter of any MPC formulation is the prediction horizon length. Too short prediction horizons may result

TABLE II

TRAINING, VALIDATION AND TESTING PERFORMANCE FOR DIFFERENT DNN SETTINGS AND DATASETS FOR APPROXIMATE EMPC

DNN setting	Data	Performance (MSE)		
		Training	Validation	Testing
L=5, M=10	500 K random	0.00260	0.00260	0.00260
L=5, M=20	500 K random	0.00053	0.00058	0.00057
L=5, M=10	1000 K random	0.00240	0.00240	0.00240
L=5, M=20	1000 K random	0.00048	0.00050	0.00051

TABLE III
FROBENIUS NORM OF DNN WEIGHTS

DNN	$\ \mathbf{W}_1\ _F$	$\ \mathbf{W}_2\ _F$	$\ \mathbf{W}_3\ _F$	$\ \mathbf{W}_4\ _F$	$\ \mathbf{W}_5\ _F$	$\ \mathbf{W}_6\ _F$
Del-MPC	1.52	14.21	13.27	15.52	17.56	3.73
EMPC	6.02	15.31	17.94	19.26	25.48	5.62

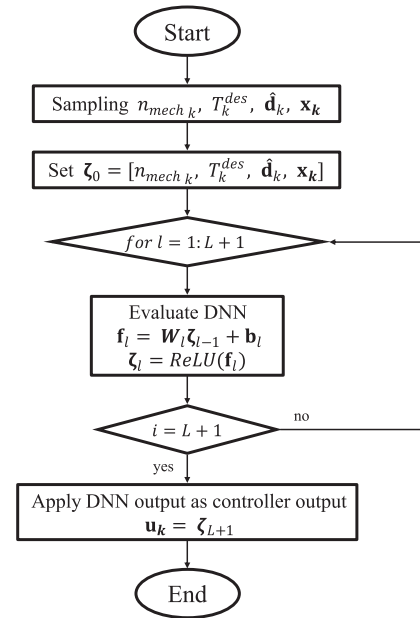


Fig. 4. Flowchart of DNN-based approximate MPC implementation.

in stability problems if not incorporated with additional terminal cost and constraints [58]. A longer horizon can lead to a better performance due to the increased freedom the optimizer has in solving the optimization problem. In practice, short prediction horizons are usually used due to limitation in computational resources of embedded systems, especially in systems with high sampling frequencies like power electronics and electric drives. Most real-time implementations of CCS-MPC for PMSMs found in the literature (shown in Table I) consider only short horizons in the range (1–5) as in [10], [14], [42], and [46]. Other implementations that consider a longer horizon of ten, as in [43] and [44], are limited to long sampling times. For the selection of the prediction horizon in this work, the benefit of longer prediction horizon on the torque's transient response is investigated. Fig. 5 shows the simulation results of a torque step from 0 to maximum torque for the EMPC formulation as an example. Three values of horizon length (2, 20, and 50)

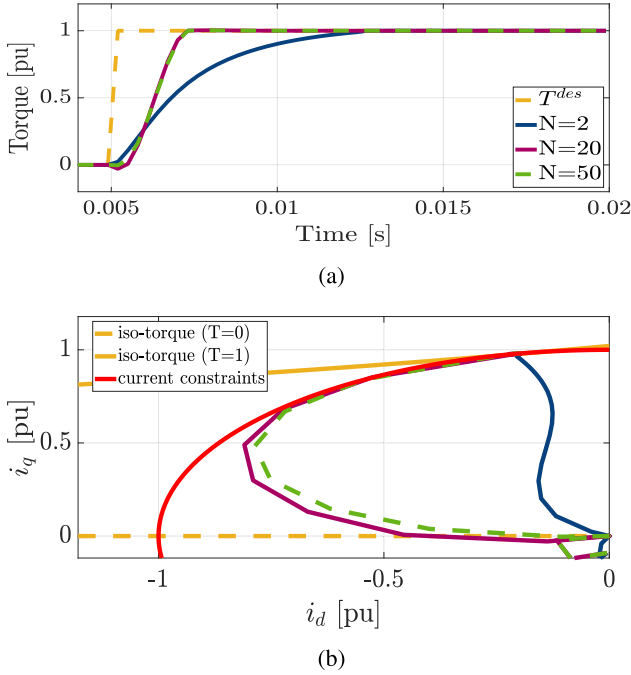


Fig. 5. Effect of horizon length of EMPC ($n_{mech} = 0.375$ p.u.) (a) Torque response. (b) Current phase diagram.

were considered, where all other MPC parameters remained unchanged. Increasing the horizon length from 2 to 20 improved the transient response significantly as can be seen in subplot (a). Further increase of the horizon to $N = 50$ resulted in no further improvement. In the current phase diagram in subplot (b), the approaches for different horizon lengths are shown, where the controller takes different currents trajectories to the new operation point. For longer horizons, smaller i_q currents are injected at first which leads to faster torque responses. The trajectory for $N = 2$ is completely different from the one for $N = 20$. The trajectory for $N = 20$ is almost identical to the one with $N = 50$. So, the horizon length $N = 20$ is considered for both MPC formulations.

C. Sensitivity to Parameter Uncertainties

Fig. 6 shows the simulation of a torque step with and without plant-model mismatch for EMPC as an example (same explanation also applies for the case of Del-MPC). The mismatch is simulated by introducing a 20% increase in the motor's resistance and 20% decrease in the dq -inductances, while keeping the nominal model's parameters used in the controller unchanged. For the case without mismatch, the desired torque is reached with no steady-state deviation and the estimated disturbance, after the Kalman filter transient, converges to zero. In the case with the introduced mismatch, the estimated disturbance by the Kalman filter converges to nonzero values. Accordingly, the controller adjusts the dq -voltage values (here, u_d and u_q are increased to account for the negative $\hat{d}_d$ and $\hat{d}_q$) to maintain offset-free torque tracking as can be seen from Fig. 6. This demonstrates that the

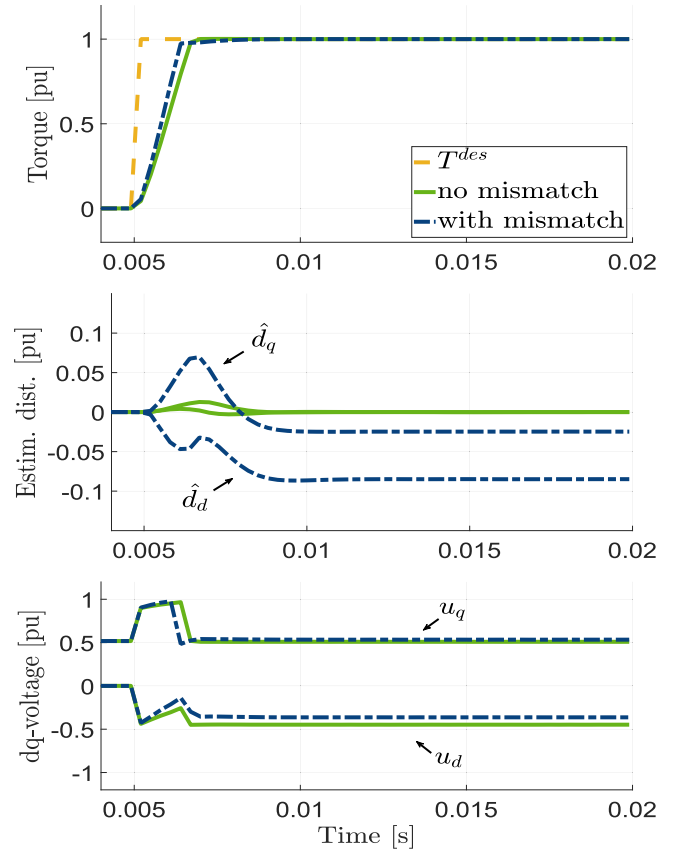


Fig. 6. Effect of plant-model mismatch (resistance: +20% and inductance: -20%) on offset-free torque tracking capability of EMPC ($n_{mech} = 0.25$ p.u.).

TABLE IV
EFFECT OF INCLUDING STEADY-STATE POINTS IN THE TRAINING SET ON TORQUE DEVIATION OF APPROXIMATE EMPC

	Operating point [p.u.]	Torque SS deviation [%]	
		without SS points	with SS points
1	$n_{mech} = 0.125, T^{des} = 0.562$	-0.03	0.01
2	$n_{mech} = 0.250, T^{des} = 0.899$	0.03	-0.17
3	$n_{mech} = 0.450, T^{des} = 0.943$	-1.56	-1.10
4	$n_{mech} = 0.475, T^{des} = 0.910$	-1.04	-0.98
5	$n_{mech} = 0.500, T^{des} = 0.876$	-1.84	-0.78
6	$n_{mech} = 0.600, T^{des} = 0.742$	-1.97	-0.45
7	$n_{mech} = 0.625, T^{des} = 0.708$	-1.40	-0.79
8	$n_{mech} = 0.500, T^{des} = 1$	-3.73	0.74
9	$n_{mech} = 0.600, T^{des} = 1$	-3.40	-0.78
10	$n_{mech} = 0.625, T^{des} = 1$	-2.40	-0.03

proposed MPC formulations are robust against model parameters inaccuracies and can achieve offset-free torque tracking.

D. DNN-Based Approximate MPC

The effect of adding steady-state (SS) focused points to the training data of the DNN-based MPC is shown in Table IV, where ten operating points from approximate EMPC are shown as an example. The tracking error due to DNN approximation is below 1.1% in the considered scenarios which is acceptable in the considered application. It can be seen that in base speed

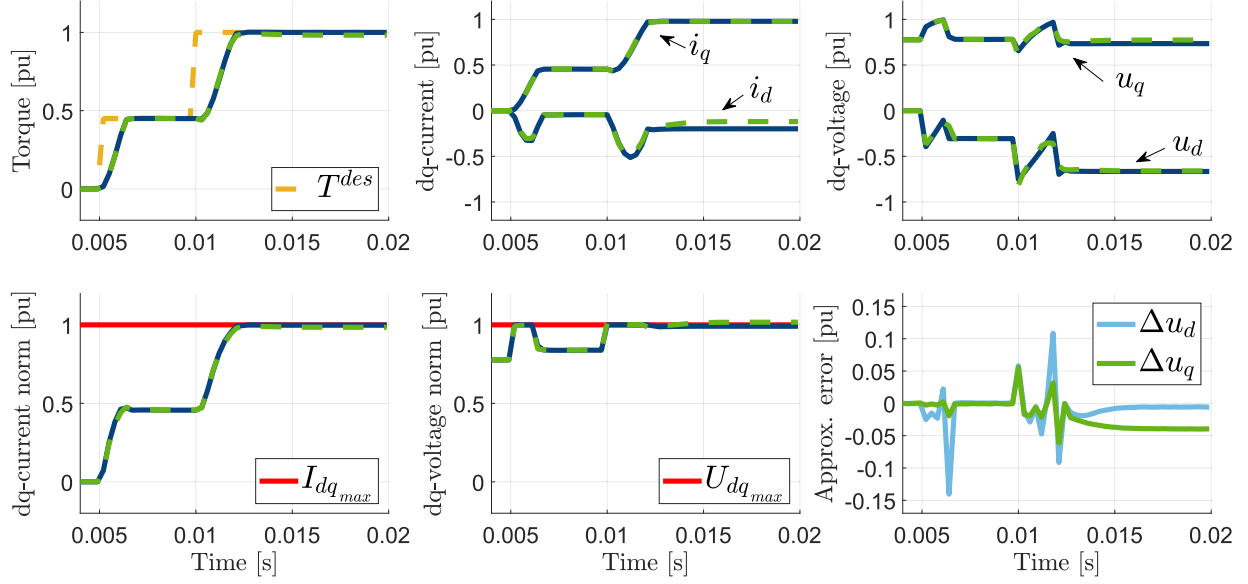


Fig. 7. Comparing exact (solid line) and approximate (dashed line) Del-MPC with two torque steps ($n_{\text{mech}} = 0.375$ p.u.).

region (operating points 1 and 2), adding SS points has very small effect on the torque deviation. In FW region (points 3–7), however, considering SS points in the training results in an obvious decrease in the torque deviation. A considerable enhancement can be seen for points with unreachable desired torque (points 7–10), where addition of SS point to the training data improves the DNN’s ability to recognize the maximum reachable torque more accurately. It should be noted that no plant-model mismatch is considered for the operating points in Table IV. For the case with high estimated disturbance due to large parameters inaccuracy, considering SS points can result in a more significant reduction in torque deviation.

Although real-time implementation of the exact MPC using the proposed formulations and long horizon is infeasible, it could be compared to the DNN-based MPC in simulation. Fig. 7 shows simulation results of two torque steps in FW region for the exact and DNN-based Del-MPC as an example. It can be seen that the approximate Del-MPC is very close to its exact counterpart. The approximate Del-MPC is capable to track the desired torque with negligible steady-state deviation, and it respects current and voltage limits. The approximation error, difference between the control voltage $\mathbf{u}_{dq}$ from the exact MPC and the one computed by the DNN, can also be seen to be very small and close to zero. It has a higher value during transient but then decreases at steady state. Since the second torque step corresponds to operating at the current and voltage limits, its approximation is generally more challenging, and hence the approximation error is slightly larger comparing with the error at the first torque step. Due to approximation error, a slight voltage limit violation occurred during the transient of the second torque step.

Generally, steady-state torque deviation and limit violations due to DNN approximation are expected from an approximate MPC implementation. However, they are still negligible in both DNN-based MPC formulations, even for the most challenging-to-approximate operating points corresponding to voltage and

current limits. For test bench implementation, voltage saturation is added to mitigate any possible voltage limit violation due to approximation error.

The average computation time (in s) needed to solve the exact MPC formulations in (6) and (7) with $N = 20$, using the IPOPT solver of the CasADi package, are 2×10^{-1} s and 5×10^{-1} s, respectively. On the other hand, the computation time of the two DNNs representing the approximate Del-MPC and EMPC is around 1.5×10^{-5} s. The simulation was performed on a personal computer with Intel XEON E3-1505 M v6. It is clear that the exact MPC needs very long time, with EMPC having considerably longer computation time than Del-MPC due to its higher complexity. This makes real-time deployment of such long horizon MPC formulations impossible, especially with a sampling period of $100 \mu\text{s}$. The DNN-based MPC, on the other hand, needs only $15 \mu\text{s}$ in average, making it easily real-time implementable, even with sampling times smaller than $100 \mu\text{s}$.

VI. EXPERIMENTAL RESULTS

The DNN-based MPCs were deployed at the test bench on the rapid prototyping system dSPACE DS1007 real-time platform with MATLAB/Simulink and control desk interface. The test bench consists of the IPMSM (described in Section V) coupled with a load machine for rotor mechanical speed control. Sampling as well as the switching frequency of 10 kHz is used. The experimental setup is shown in Fig. 8. As mentioned earlier, the Kalman filter is used for disturbance estimation, with computation time requirement of around $1.64 \mu\text{s}$.

In this section, experimental results of the DNN-based Del-MPC and EMPC are presented. The general torque tracking performance is shown first and the DNN capability to recognize infeasible desired torque and to comply with limits is demonstrated, then comparisons with the state-of-the-art FOC using



Fig. 8. Experimental setup utilized to validate the proposed DNN-based MPC.

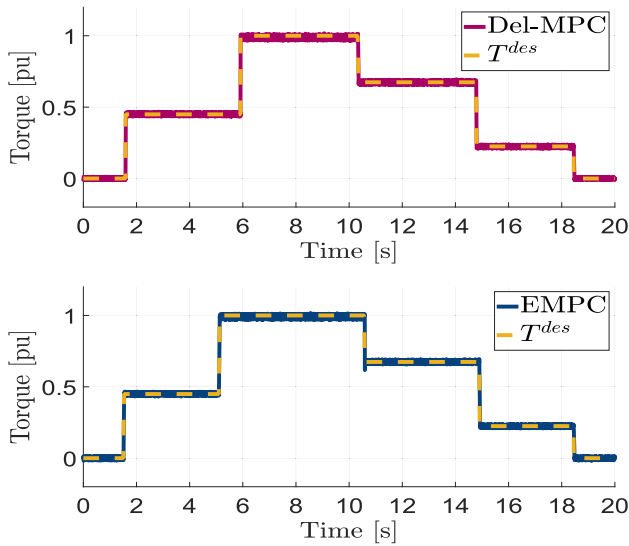


Fig. 9. Torque tracking performance of DNN-approximate Del-MPC and EMPC ($n_{\text{mech}} = 0.25$ p.u.).

IMC method, and with the exact (online solved) short-horizon MPC using FGM are presented.

A. Torque Tracking Performance

Fig. 9 shows the torque tracking performance of the DNN-based approximate Del-MPC and EMPC for a series of ascending and descending desired torque profile covering the range from 0 to maximum torque at a speed of (0.25 p.u.). The torque shown is the estimated electromagnetic torque from the measured current. Both approximate Del-MPC and EMPC can be seen to have a very good torque tracking capability without steady-state offset.

B. Compliance With Limits

Since the DNN replaces not only the MPC function, but also the reference generator function, it should be able to recognize infeasible desired torques at any operating point, which are unknown in advance since it depends on the estimated disturbance. So, it should be able to recognize the maximum possible torque

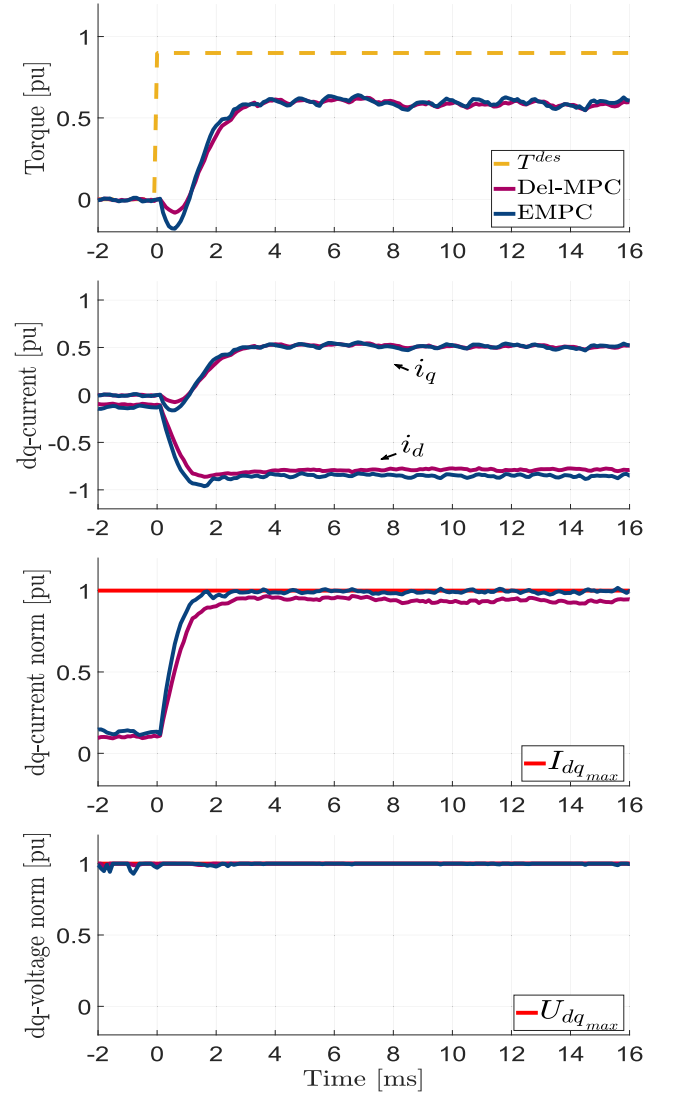


Fig. 10. DNN-approximate MPC performance and limits compliance at unreachable desired torque ($n_{\text{mech}} = 0.5$ p.u.).

that can be tracked at any operating point while complying with the current and voltage limits, and without the need for any LUTs to store the speed-torque characteristic of the motor or the need for a field-weakening controller, as in conventional FOC.

Fig. 10 shows the performance of approximate Del-MPC and EMPC at a relatively high speed of (0.5 p.u.) with an unreachable desired torque, which is directly input to the DNNs. Both approximate MPCs recognize this and control the motor to produce the maximum reachable torque corresponding to the motor's current limitation. Due to complexity of EMPC formulation, DNN approximation error may result in negligible slight current limit violation. Any violation in voltage limit due to approximation is mitigated with a voltage saturation.

It should be noted that due to manufacturing imperfections of the motor, slight harmonic disturbances (dominated by the sixth electrical harmonic) exist in the measured current, and hence, in the estimated torque. This is, however, unrelated to the MPC performance.

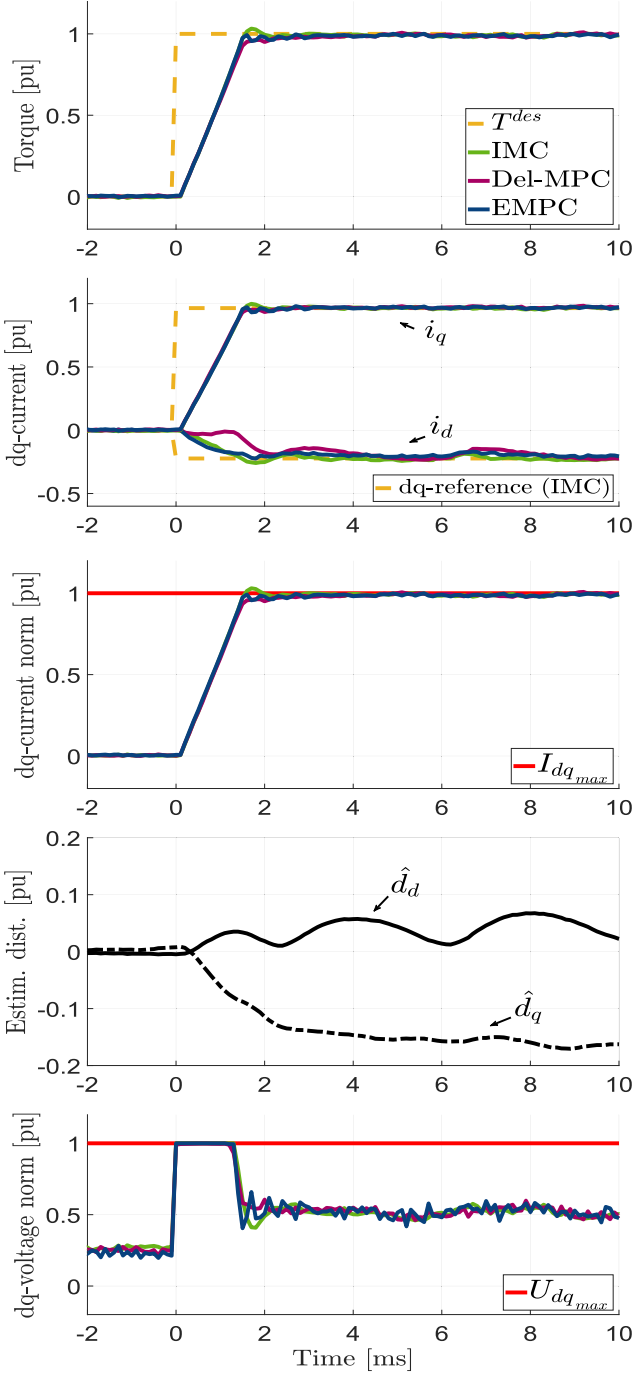


Fig. 11. Comparison between DNN-based Del-MPC and EMPC with IMC in base speed region ($n_{\text{mech}} = 0.125$ p.u.).

C. Comparison to State-of-The-Art Controllers

Performance of the DNN-based approximate Del-MPC and EMPC is first compared with the state-of-the-art FOC based on IMC method. It should be noted that the IMC was tuned manually to show good torque tracking behavior in the whole operation region.

Fig. 11 shows the comparison between IMC and approximate Del-MPC and EMPC with a torque step (0–1 p.u.) at a low speed

TABLE V
RISE TIME OF TORQUE SIGNAL IN FIG. 12

Controller	$t_{0-50\%}^{\text{rise}}$ [ms]	$t_{0-90\%}^{\text{rise}}$ [ms]
IMC+FW	2.9	3.7
Del-MPC	1.8	3.1
EMPC	1.5	3.0

($n_{\text{mech}} = 0.125$ p.u.). As can be seen, all three controllers have a comparable torque transient behavior, although the IMC has a small torque and current overshoot. The shown reference dq-currents are only required for the IMC and were obtained from an LUT generated by solving an offline optimization problem. DNN-based Del-MPC and EMPC do not require any reference currents for torque tracking, since they have already learned the reference generator function during their training. All controllers apply the maximum voltage during transient, as can be seen from the dq -voltage norm, before they converge to the new steady-state operating point. One may also note that IMC and Del-MPC have a smoother voltage signal than EMPC. This is due to the current tracking nature of both IMC and Del-MPC in comparison with torque tracking nature of EMPC.

The comparison shown in Fig. 12 corresponds to a torque step (0.45–0.67 p.u.) at a speed ($n_{\text{mech}} = 0.375$ p.u.), where both operating points lie on the voltage limit. A considerable difference in the torque transient can be seen, with both Del-MPC and EMPC having faster torque transients in comparison to IMC with FW part, even with allowed overshoot. As can be seen from rising times of the three controllers in Table V, both Del-MPC and EMPC required considerably less time than IMC+FW to reach 50% of the desired torque. Due to the dynamics of the Kalman-filter used for disturbance estimation, the torque transient of Del-MPC and EMPC slows down slightly when approaching the reference. This should be mitigated with fine tuning of the Kalman-filter and is not considered here. Nevertheless, both Del-MPC and EMPC are still faster than IMC+FW in reaching 90% of the desired torque. Due to operation in the field-weakening region, a FW part is required in case of IMC for online adjustment of the reference dq -currents to maintain operation at voltage limit. This process requires some time until convergence to the new steady-state point is achieved, as can be seen in the d -reference current, which has not completely converged to its steady-state value, even after 14 ms. This leads to a higher amplitude in current, and thus, to higher copper losses in the machine during transient.

A comparison at another operating point in the FW region is shown in Fig. 13, where a torque step (0–0.225 p.u.) is applied at a higher speed ($n_{\text{mech}} = 0.6$ p.u.). Here, also a considerable difference in the torque transient can be seen, with IMC having slower transient than both approximate Del-MPC and EMPC. Despite the effect of Kalman-filter's dynamics, both approximate Del-MPC and EMPC have a 0–90% rising time that is 44% less in comparison with IMC+FW, as can be seen in Table VI. Online adjustment of IMC reference currents, due to operating at voltage limit, can also be seen in the dq -current plot in Fig. 13. Both DNN-based approximate MPCs managed to recognize

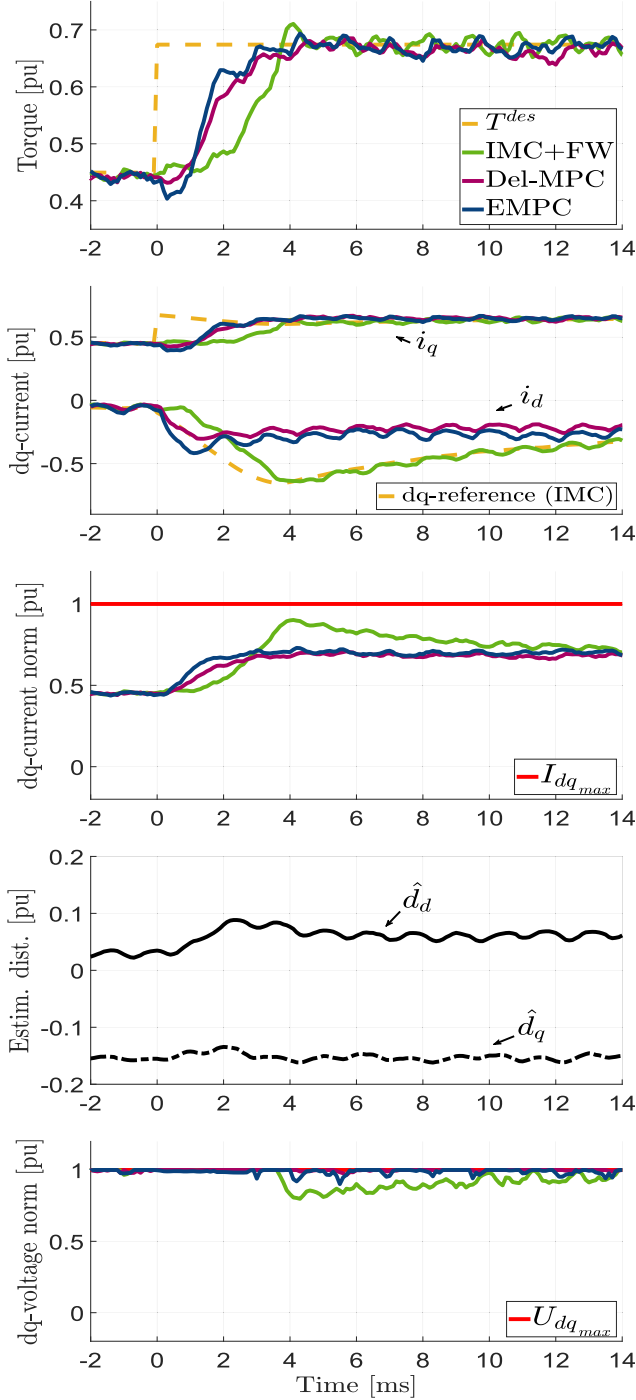


Fig. 12. Comparison between DNN-based Del-MPC and EMPC with IMC in FW region ($n_{\text{mech}} = 0.375$ p.u.).

TABLE VI
RISE TIME OF TORQUE SIGNAL IN FIG. 13

Controller	$t_{0-50\%}^{\text{rise}}$ [ms]	$t_{0-90\%}^{\text{rise}}$ [ms]
IMC+FW	1.7	2.5
Del-MPC	0.9	1.4
EMPC	0.9	1.4

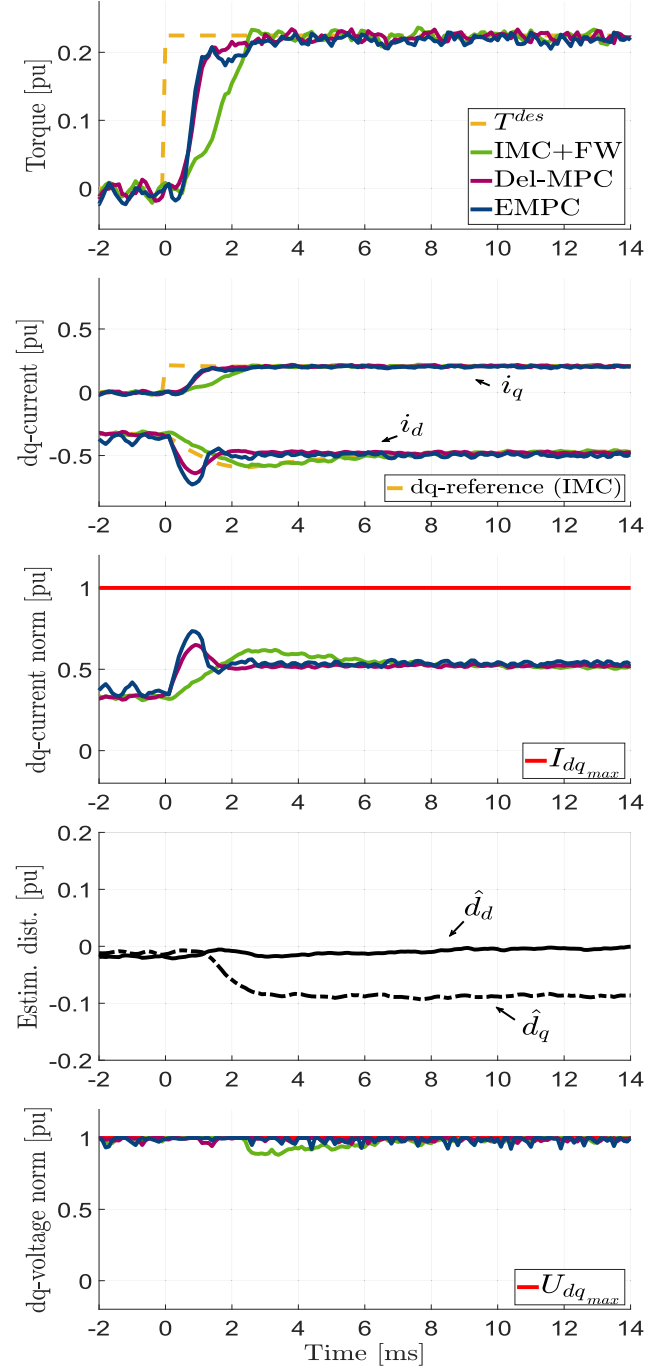


Fig. 13. Comparison between DNN-based Del-MPC and EMPC with IMC in FW region ($n_{\text{mech}} = 0.6$ p.u.).

and to converge much faster to the new steady-state dq -current values, without the need for an explicit current reference.

So, generally when operating away from the voltage limit, as in base-speed region, no much difference can be seen in the transient performance of approximate MPC in comparison with conventional IMC. In the field-weakening region, however, both approximate MPCs have inherited the reference generator capabilities to determine the new steady-state current operating values corresponding to a desired torque, and hence manage

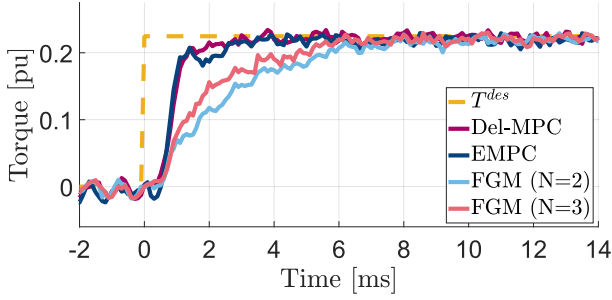


Fig. 14. Comparison of torque response between DNN-approximate long-horizon Del-MPC and EMPC with exact short-horizon FGM-MPC in FW region ($n_{\text{mech}} = 0.6$ p.u.).

TABLE VII
COMPUTATION TIME OF CONTROLLERS IN FIG. 14

Controller	Computation time [μs]
DNN-based Del-MPC ($N = 20$)	10.88
DNN-based EMPC ($N = 20$)	10.81
FGM-MPC ($N = 2$)	26.05
FGM-MPC ($N = 3$)	37.99

to converge to them much faster than IMC, which depends on the FW part for online adjustment of reference currents. This results in Del-MPC and EMPC having generally faster torque transients. Due to the direct consideration of torque in the stage cost function of EMPC, it could, at some operating points, achieve faster torque transients in comparison with the current tracking Del-MPC as seen from Fig. 12 and Table V.

It should be noted that a significant plant-model mismatch exists at the test bench in comparison with the motor's nominal model, used for both MPCs and in the conventional IMC, as can be seen from the estimated disturbance ($\hat{d}_d$ and $\hat{d}_q$) in Figs. 11–13. This demonstrates the robustness of the proposed Del-MPC and EMPC against model uncertainties. The conventional IMC is also robust against this plant-model mismatch through its use of the integral action.

The comparison with the exact (online-solved) short-horizon MPC using FGM is shown in Fig. 14, where two short horizons are considered ($N = 2$ and $N = 3$). As can be seen, FGM-MPC with ($N = 3$) results in a faster torque transient than the one with ($N = 2$), which demonstrates again the benefit of increasing the horizon length on control performance. However, both FGM-MPCs are still significantly slower than the DNN-based long-horizon Del-MPC and EMPC due to their short horizons. This is despite their significantly longer computation time in comparison with the DNN-based MPCs, as can be seen from Table VII. This demonstrates the benefit of the proposed approach of approximating long-horizon MPC through computationally efficient DNNs in improving the control performance of PMSM, especially at voltage and current constraints.

VII. CONCLUSION

In this article, two online-efficient and high performing approximate long-horizon CCS-MPC formulations were presented

for enhancing the control performance of PMSM. The classical current tracking Del-MPC tracks the steady-state reference dq -currents and voltages, corresponding to a desired torque, generated by a generic reference generator. The torque tracking EMPC considers the torque deviation directly in the stage cost through a slack variable instead of current and voltage tracking terms. Estimated plant-model mismatch as input disturbance is considered to maintain operation within current and voltage limits, ensuring offset-free torque tracking under model uncertainties.

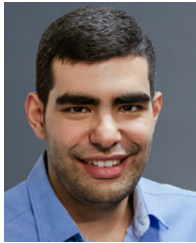
To enable real-time implementation of the long-horizon MPC, approximation through DNNs is used, where the exact (ground truth) MPC is sampled to generate data (offline and in open loop) needed to train the corresponding DNN that will be evaluated at run time. This approach requires much less effort, can be entirely automated, and ensures an online-efficient approximation of long-horizon MPC, in comparison with the development of online-capable solvers that can even be infeasible for long horizons.

The DNN-based long-horizon approximate MPC was compared in simulation to the exact MPC and the results illustrate a very good approximation performance with significantly lower computation time requirement. Validation through test bench implementation verified its robustness and offset-free torque tracking capability. A comparison to the state-of-the-art IMC-based FOC showed a considerable enhancement in the torque transient response of both approximate MPCs, mainly in the FW region, with a decrease in torque rising time that can reach up to 44%. A comparison with the exact short-horizon FGM-MPC demonstrated the effect of horizon length on control performance, where a significantly faster torque transient response is achieved by the DNN-based long-horizon MPC.

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